BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 191 · Procedures, functions and parameters

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. A local x=4 shadows a field x=9. Which is printed inside the method?

Q2. What is a base case?

Q3. What is a[2] for int[] a = {3,5,8,13}?

Q4. Starting x=1, apply x=x*2 three times. List the results.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

procedure — A subroutine that performs actions and does not return a result.

function — A subroutine that returns a value.

parameter — A named input in the subroutine declaration.

argument — The actual value provided by the caller.

return — Finish a function and pass its result to the caller.

New learning

CALL — The caller supplies argument 6 to Double.

PARAMETER — Parameter value receives 6, then value * 2 evaluates to 12.

RETURN — The function returns 12. Console.WriteLine displays the result.

Worked example

Trace AddVAT(50), which returns price + price / 5.

1. Predict first — The argument 50 becomes local parameter price = 50.

2. Model the result — Integer division 50 / 5 = 10; the returned value is 60.

We Do · Practise with guidance

What is printed by TriplePlusOne(4)?

```
static int TriplePlusOne(int n)
{
    return 3 * n + 1;
}
Console.WriteLine(TriplePlusOne(4));
```

Predict first

The argument 4 is assigned to n.

Model the result

3 × 4 + 1 = 13, so the method returns 13 and prints 13.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

A function Square(n) returns n*n. Find Square(5).

Question 2

Write a C# function IsEven(int n) that returns true when n is even.

Question 3

Explain why static int Show(int x) { Console.WriteLine(x); } is invalid. Correct it.

Exit Ticket · Answer before checking

A C# method with return type void must return an integer.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 4. The local variable shadows the field.

Q2. The condition that stops further recursive calls.

Q3. 8, because arrays start at index zero.

Q4. 2, 4, 8.

You Do pages · Independent questions

Q1. 25.

Q2. static bool IsEven(int n) { return n % 2 == 0; }

Q3. The int return type requires an int result. Use static void Show(int x) { Console.WriteLine(x); } or return x and print at the call site.

Exit Ticket · Model answer

False. A void method does not return a result.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: